

NexusPortal: Development Blueprint

Full-Stack Self-Study Command Center & Learning Ecosystem

Project Overview

NexusPortal is a specialized Learning Management System (LMS) designed for adult learners. It integrates curriculum tracking, personal productivity tools, and interactive document review into a unified dashboard.

Technical Architecture

Frontend	HTML5, CSS3 (Tailwind), Vanilla JavaScript (AJAX)
Backend	Python / Django (MVT Architecture)
Database	SQLite (Relational Storage)
Core Features	Course Progress, Document Vault, Pomodoro Timer, Today's To-Do

HTML Mastery: 25-Minute Sprint

Phase 1: Semantic Structure (10m)

Implementation of semantic tags (<main>, <section>, <article>) to ensure document accessibility and SEO optimization.

Phase 2: Accessible Forms (10m)

Developing robust data entry points using <label> and <input> associations, crucial for the 'New Path' and 'Upload Resource' modules.

Phase 3: Metadata & SEO (5m)

Configuration of viewport settings and Open Graph tags to ensure responsive rendering and professional link sharing.

Next Implementation Steps

- **Smart Calendar:** Integration of FullCalendar.js for automated study scheduling.
- **PDF Analytics:** Developing the split-screen PDF viewer with an 'Active Recall' note-taking pane.
- **Spaced Repetition:** Implementing a Leitner System algorithm for long-term knowledge retention.